5.8 Importance and applications of coordination complexes:

The coordination complexes are of great importance. These compounds are present in many plants, animals and in minerals. Some Important applications of coordination complexes are described below.

1. Phthalo blue – a bright blue pigment is a complex of Copper (II) ion and it is used in printing ink and in the packaging industry.
2. Purification of Nickel by Mond's process involves formation $[Ni(CO)_4]$, which yields 99.5% pure Nickel on decomposition.
3. EDTA is used as a chelating ligand for the separation of lanthanides,in softening of hard water and also in removing lead poisoning.
4. Coordination complexes are used in the extraction of silver and gold from their ores by forming soluble cyano complex. These cyano complexes are reduced by zinc to yield metals. This process is called as Mac-Arthur –Forrest cyanide process.
5. Some metal ions are estimated more accurately by complex formation. For example, Ni^{2+} ions present in Nickel chloride solution is estimated accurately for forming an insoluble complex called $[Ni(DMG)_2]$.
6. Many of the complexes are used as catalysts in organic and inorganic reactions. For example,
   (i) Wilkinson's catalyst - $\left[(PPh_3)_3\,RhCl\right]$ is used for hydrogenation of alkenes.
   (ii) Ziegler-Natta catalyst - $[TiCl_4\ ]+\ Al(C_2H_5)_3$ is used in the polymerization of ethene.
7. In order to get a fine and uniform deposit of superior metals (Ag, Au, Pt etc.,) over base metals, Coordination complexes $\left[Ag(CN)_2\right]^-$ and $\left[Au(CN)_2\right]^-$ etc., are used in electrolytic bath.
8. Many complexes are used as medicines for the treatment of various diseases. For example,
   (1) Ca-EDTA chelate, is used in the treatment of lead and radioactive poisoning. That is for removing lead and radioactive metal ions from the body.
   (2) Cis-platin is used as an antitumor drug in cancer treatment.
9. In photography, when the developed film is washed with sodium thio sulphatesolution (hypo), the negative film gets fixed. Undecomposed AgBr forms a soluble complex called sodiumdithiosulphatoargentate(I) which can be easily removed by washing the film with water.

$$AgBr\ +\ 2\ Na_2S_2O_3 \longrightarrow Na_3\left[Ag(S_2O_3)_2\right]\ +\ 2\ NaBr$$

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